

## **Group 8D - Vacuum Robot**

### **Power/energy requirement**

- Source voltage: **12 V nominal**
- Average current at nominal source voltage: **~1.0 A**
- Maximum/peak current at nominal source voltage: **~2.5 A**
- Required operating time: **10–20 min**
- Estimated energy required: **2–4 Wh**

### **Constraints (Battery)**

- Maximum size: **15 cm length × 10 cm width**
- Maximum weight: **700 g**
- Other constraints: The battery should be capable of supplying the estimated peak current without exceeding a 1C discharge rate. The project is designed around a nominal 12 V supply, with a 12.8 V nominal LiFePO4 battery pack considered acceptable pending verification of the electronics' voltage requirements. The battery should fit within the available physical space of the robot and remain below the maximum weight limit.

### **Status (Battery)**

- Discussed with supervisors: **No**
- Battery selected: **Yes — preliminary selection**
- Battery approved: **Pending**

### **Proposed solution (Battery)**

- Nominal voltage: **12.8 V**
- Capacity / energy: **3 Ah / ~38.4 Wh**
- Configuration: **2 × 6.4 V, 3000 mAh LiFePO4 batteries connected in series**
- Product link:  
**<https://www.batteribyen.dk/batterier/industribatterier/nodbelysning/batteri-til-n-dbelysning-dotlux-6-4v-3000mah->**

### **Safety considerations (Battery)**

- Battery switch / main disconnect: **Yes**
- Fuse: **Yes — recommended**
- BMS/protection: **No**
- Other safety considerations: A temperature sensor and a current sensor are used to monitor the battery temperature and the current being drawn from the battery which can provide information for identifying potentially unsafe operating conditions.

*Calculations of peak current and average current below.*

| ID  | Purpose                        | Name                             | Price per Unit(dkk) | Quantity | Reseller         | Current (A)                                                  |
|-----|--------------------------------|----------------------------------|---------------------|----------|------------------|--------------------------------------------------------------|
| C1  | Drive Motors                   | Nidec DC Gear Motor              | 196                 | 2        | RS               | Stalling: 0.6<br><br>Read from graph at approx. 65 RPM: 0.12 |
| C2  | Vacuum Blower                  | Sanyo Denki Blister              | 190                 | 1        | RS               | 0.28                                                         |
| C2  | Vacuum Blower Driver           | L298N                            | 35                  | 1        | Mechatronics Lab | negligible                                                   |
| C3  | Brush Motor                    | DC Micro Motor                   | 15                  | 1        | Ardustore        | 0.1                                                          |
| C4  | Buck Boost Brush Motor         | Auto Buck Boost Converter Module | 30                  | 1        | Ardustore        | negligible                                                   |
| C5  | Microcontroller                | Seeit ESP32-DEV-38P              | 126                 | 1        | RS               | 0.1 to 0.3                                                   |
| C6  | Motordriver Drive              | DFRobot TB6612FNG                | 29                  | 1        | Farnell          | negligible                                                   |
| C7  | Heat Recognition               | LM75                             | 27                  | 1        | Mechatronics Lab | negligible                                                   |
| C8  | Battery Capacity               | TSC214ICT                        | 8                   | 1        | Farnell          | negligible                                                   |
| C9  | Collector Indicator            | Optek Infrared Sensor            | 60                  | 1        | RS               | 0.1                                                          |
| C10 | Wheels                         | Mecanum Wheel                    | 84                  | 2        | Let elektronik   |                                                              |
| C11 | Ball Wheel                     | Ball Wheel                       | 22.5                | 1        | Let elektronik   |                                                              |
| C12 | Navigation Sensor              | VL53L1X ToF Sensor               | 75                  | 1        | Ardustore        | negligible                                                   |
| C13 | Navigation Sensor Servo        | DFRobot DF9GMS                   | 32                  | 1        | Farnell          | 0.1 to 0.2                                                   |
| C14 | Navigation Sensor Side Sensors | Sharp Reflektiv optisk Sensor    | 75                  | 1        | RS               | negligible                                                   |
| C15 | Gyroscope and Acceleromter     | MIKRO ELECTRONICS - BMI 270      | 175                 | 1        | RS               | negligible                                                   |

We are going to assume that the sum of negligible currents should amount to 100 mA

In the worst case scenario, Currents can reach up to 2.5A if the motors are stalling and if the microcontroller is drawing a lot of current due to overwhelming activities. In the average scenario the circuit is going to draw around 1A

